Assignment: Binary Classification Algorithms - Theory and Application

Objective: This assignment aims to deepen your understanding of fundamental binary classification algorithms, including Logistic Regression, Random Forests, and K-Nearest Neighbors. You will explore their underlying principles, mathematical formulations, and practical implications, drawing upon the concepts discussed in our course materials.

Part 1: Logistic Regression - The Probabilistic Classifier

Logistic Regression is a cornerstone algorithm for binary classification. This section explores its mathematical foundation and probabilistic interpretation.

The Sigmoid Function:

Based on the provided context, define the sigmoid function, $g(z)$, and explain its role in Logistic Regression. Why is it also commonly referred to as the logistic function?

Demonstrate mathematically how the sigmoid function satisfies the property $g(z) + g(-z) = 1$. What is the significance of this property for defining a probability model in binary classification?

Probabilistic Interpretation:

Using the sigmoid function, explain how Logistic Regression models the probability $p(Y = y | x; \theta)$ for a binary label $y \in {-1, 1}$.

Discuss how the "margin" $yx^T\theta$ influences this probability. What does it mean for the predicted probability when $yx^T\theta$ is large and positive (e.g., > 5) versus when it is quite negative?

Loss Function and Optimization:

The context presents the log-likelihood function $\ell(\theta)$ and the logistic loss function $J(\theta)$. Explain the relationship between maximizing the log-likelihood and minimizing the average logistic loss.

Write down the log-likelihood function $\ell(\theta)$ as presented in the context (Equation 2 or the expanded form for $y \in {-1, 1}$), and briefly explain why maximizing this function helps in finding the optimal parameters θ.

Part 2: Random Forests - Ensemble Power

Random Forests leverage the power of ensemble learning to achieve robust classification.

Core Concept:

Describe the fundamental principle behind Random Forests. How do they combine a "multitude of decision trees" to make a final classification decision?

Variance Reduction:

Explain the role of "bootstrapping" in the construction of Random Forests. How does bootstrapping, along with the selection of a subset of features at each split point, contribute to mitigating variance compared to using a single decision tree?

Advantages:

Based on the context, what is a key advantage of Random Forests when compared to an individual decision tree?

Part 3: K-Nearest Neighbors (KNN) - Instance-Based Learning

K-Nearest Neighbors is a simple yet effective instance-based learning algorithm.

Classification Mechanism:

How does the K-Nearest Neighbors algorithm classify a new instance $x^{(i)}$? What is the role of "majority vote" in this process?

The Parameter K:

Briefly discuss the significance of the parameter K in the K-Nearest Neighbors algorithm. How might the choice of K impact the classification outcome?

Part 4: Comparative Analysis and Application (Optional/Advanced)

This section encourages you to think critically about when and why to choose a particular algorithm.

Model Selection Considerations:

Consider a binary classification task where you need to choose between Logistic Regression, Random Forests, and K-Nearest Neighbors. Based on the descriptions provided, what are some potential strengths and weaknesses of each algorithm that might influence your choice? (e.g., interpretability, handling non-linearity, computational efficiency, sensitivity to feature scaling or outliers).

Practical Task (Conceptual):

Imagine you are tasked with building a binary classifier to predict whether a customer will "churn" (i.e., discontinue their service) or not. Briefly outline how you would approach this problem using one of the algorithms discussed (choose one: Logistic Regression, Random Forests, or KNN). Your outline should conceptually mention key steps like data preparation (e.g., what kind of features might be relevant), model training, and how you would evaluate its performance.